Experiments : Study of LAN Topology, Network Device and Address Configuration.

Problem Statement

1. Draw the Network Topology :

- Draw and identify LAN topology of your organisation and label all components such as Router, Switch, Access Point and end devices.

→ Below is detail case study of compus network.

- Layer 3 core switches in server room.
- Managed by POE + switch.
- Wireless Access point
- Multiple Ethernet port in class & lab.

* Network Topology

Hybrid Hierarchical topology which combines

- Tree topology
- Extended star topology
- Ring topology (server room)

→ Network Hierarchical layer

i> core layer - It contains server room and server firewall, Router, 2 - L3+ switches, Server, inverter etc.

ii> Distribution layer - Here, it contains 4 switches

(i) Main Build → R7 → (2nd floor)
(ii) Main Build → R1 → (3rd floor)
(iii) Main Build → R6 → (1st floor)
(iv) Main Build → R2 → (2nd floor)

And, each contains 4 switches:-

(i) 18-port Gigabit easy smart switch with 16-port. TL-SG1218MPE

(ii) SG3428XMP - 24-port Gigabit & 4-port 10GESEP+, L2+ Managed switches with 24 ports.

* SG1218MPE → 18-port gigabit "Easy smart" switch. Supports 16 port.
→ Used for CCTV, wifi etc.

* SG3428XMP → 24 POE+ port uplinks, 10G SEP, usually act as main distribution point for each floor.

iii> Access layer → It consists of 3-switches.

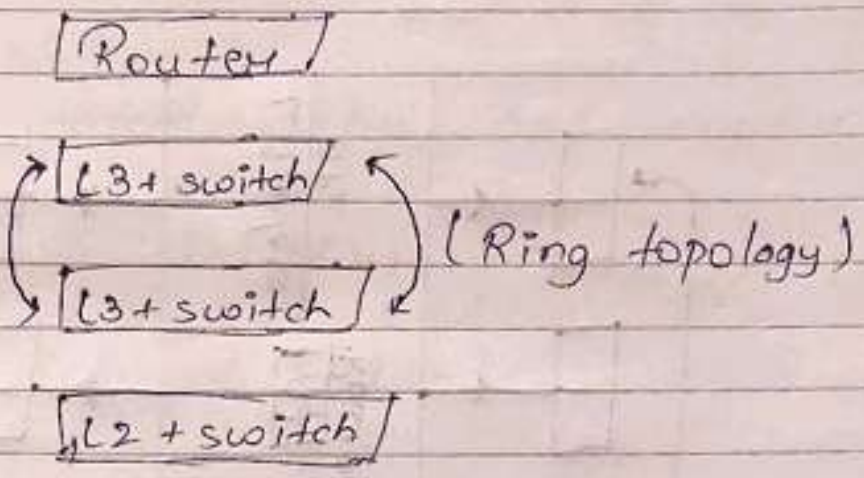
(i) R1 [Room-301, 3rd floor]
(ii) R4 [Room-05, grd floor]
(iii) R5 [Room-18, grd floor]

→ All these switches act as access layer which help to provides network connectivity to all room, lab, etc.

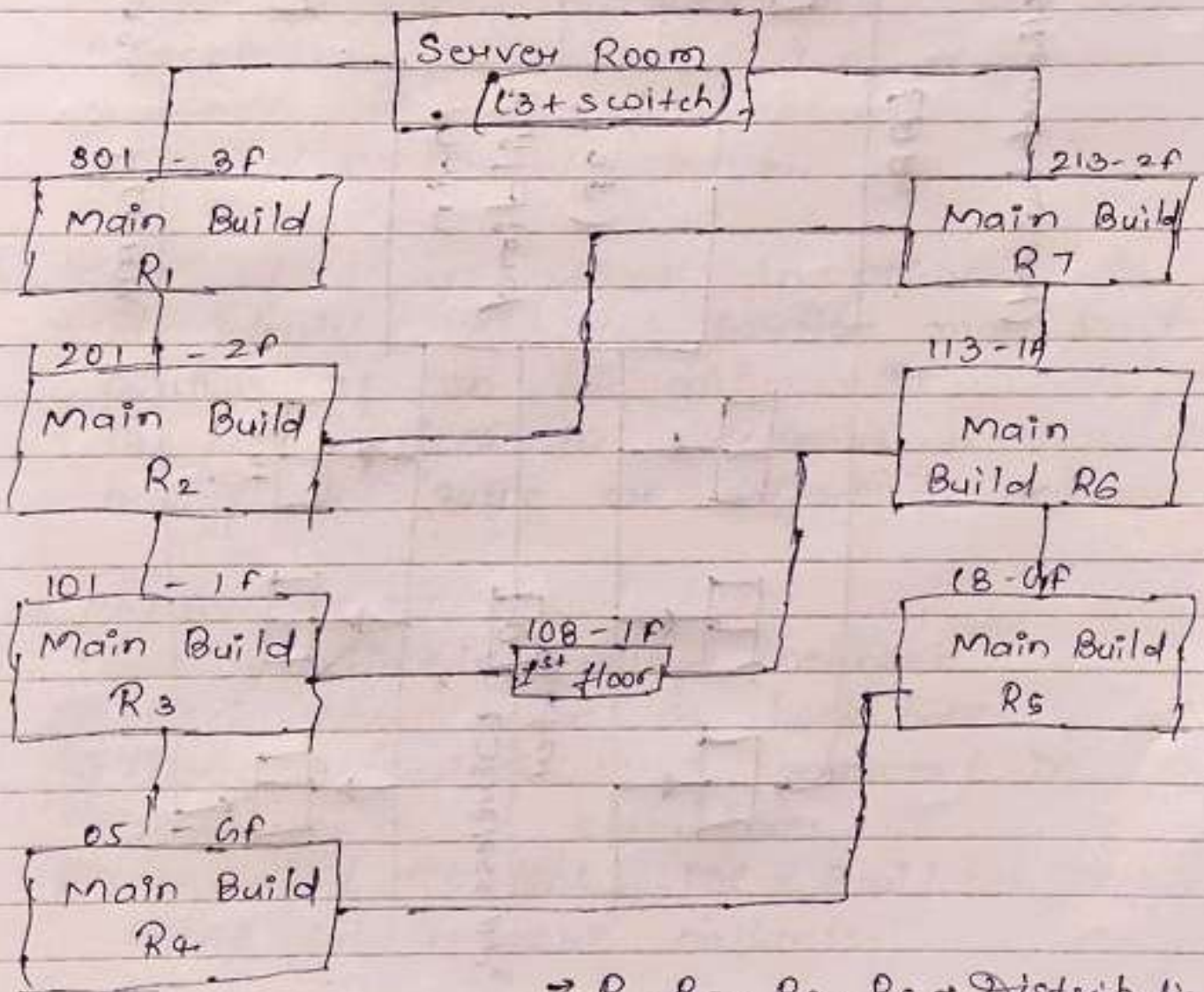
* It's directly connected to:-

(i) Ethernet (RJ45)
(ii) Wireless
(iii) Lab system
(iv) Outdoor wireless.

* Server Room Architecture / Topology

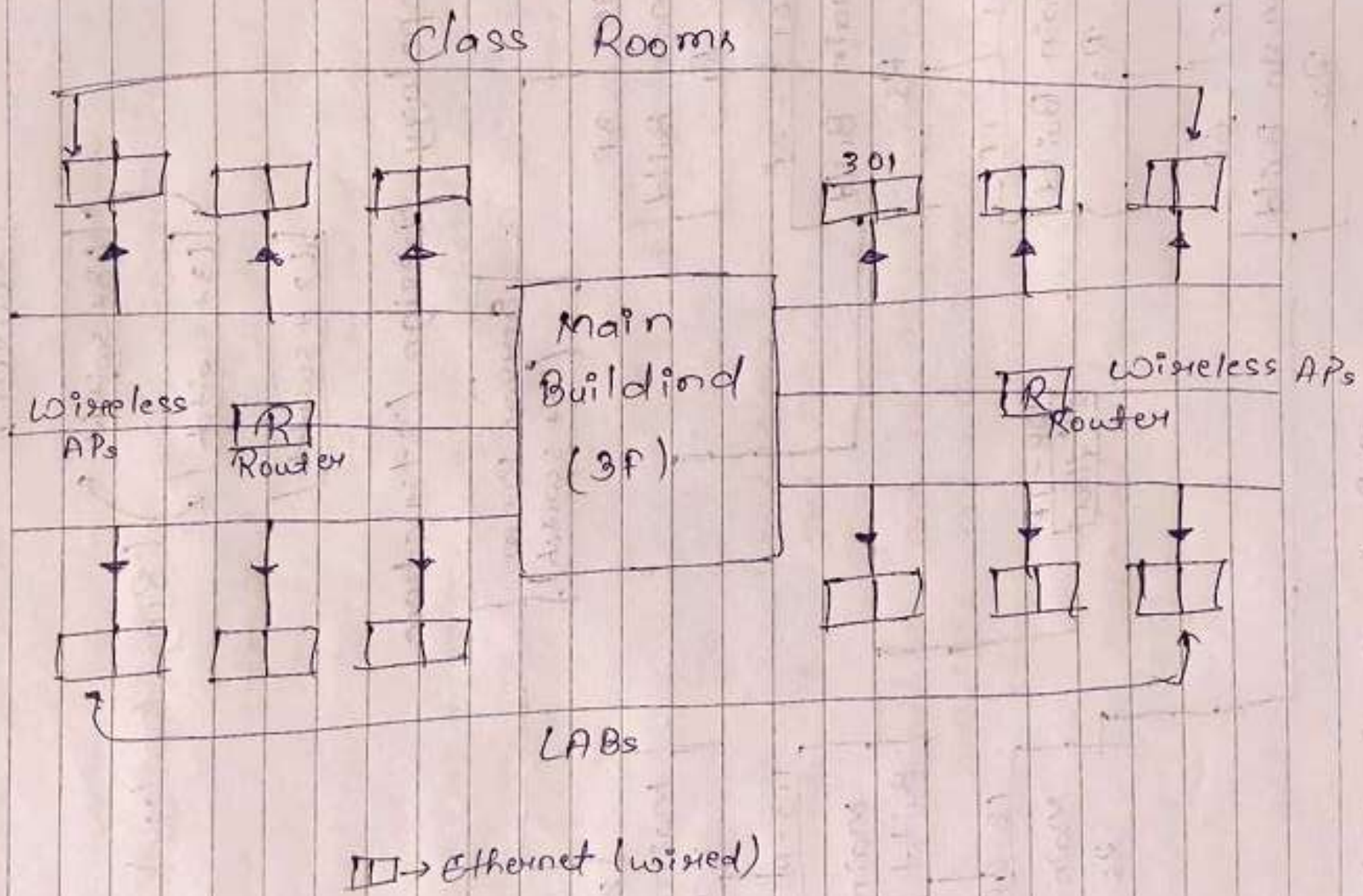


* Topology Design / structure



→ R, R7, R2, R6 → Distribution layer
→ Server room → Core layer
→ R1 — R7 → Access layer

* Access layer Switch (3rd floor - 301)



Q. Difference between wired and wireless speed?

→ Wired Network
- faster communication and connection
- Stable connection.
- Very - low latency.
- Direct / physical medium.
- Used - Ethernet (RJ45)

→ Wireless Network
- Slower connection / communication.
- Affected by distance.
- Higher latency.
- Shared medium - radio.
- Used - Router to devices.

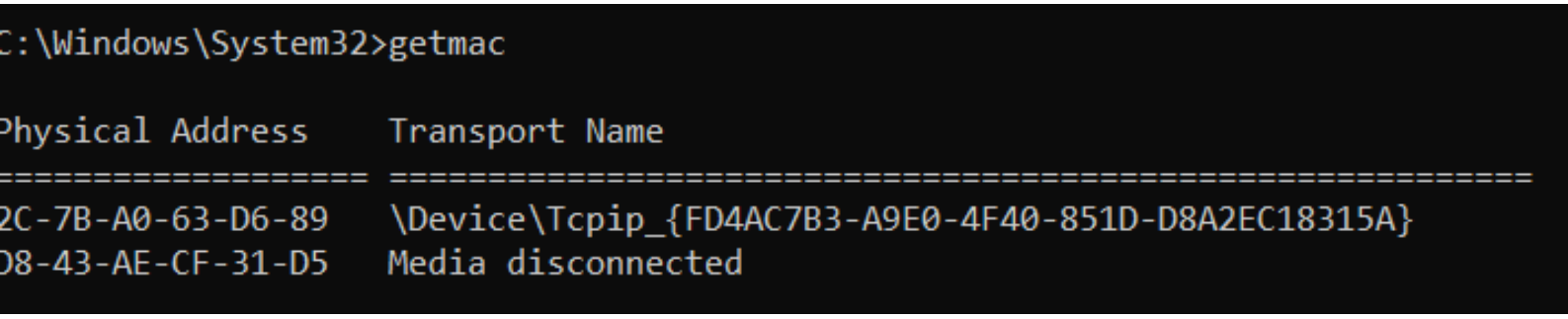
Q. Role of Subnet Mask in network.
→ It identify network portion and host portion of an IP address. It
→ It help devices to determine wether, they belong to same or different network.

* Observation :
→ During studying college network
(i) Used server room as back bone.
(ii) Two L3 switch were connected in reduced cyclic structure.
(iii) Managed TP-Link POE + Switches were used in network cabinets.

* Advantage of Existing Network distribution:
- Scalable Architecture
- Centralized managed
- Support POE+ devices
- Easy expansion of floor & class room.
- Support both wired and wireless.

* Conclusion
The campus CAN follows a hybrid hierarchical topology consists of tree, Extended star Ring. The network used to managed by POE switches wired APs. Ethernet & centralized server infrastructure and efficient wired and wireless communication across the campus.

using "getmac"



using "ipconfig"

```
C:\Windows\System32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:2018:d41e:fae0:71a6:e554:fe1d
   Temporary IPv6 Address. . . . . . : 2409:40e4:2018:d41e:5c1c:9b41:64b4:1c5
   Link-local IPv6 Address . . . . . : fe80::5101:81d5:206f:e757%20
   IPv4 Address. . . . . . . . . . . : 10.64.186.9
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::78a5:c7ff:fe14:672%20
                                       10.64.186.229
```

using "netsh wlan show inetrface"

```
C:\Users\premp>netsh wlan show interface

There is 1 interface on the system:

    Name                   : WiFi
    Description            : Intel(R) Wi-Fi 6E AX211 160MHz
    GUID                   : fd4ac7b3-a9e0-4f40-851d-d8a2ec18315a
    Physical address       : 2c:7b:a0:63:d6:89
    Interface type         : Primary
    State                  : connected
    SSID                   : POCO X6 Neo 5G
    AP BSSID               : 7a:a5:c7:14:06:72
    Band                   : 2.4 GHz
    Channel                : 11
    Connected Akm-cipher   : [ akm = 00-0f-ac:08, cipher =  00-0f-ac:04 ]
    Network type           : Infrastructure
    Radio type             : 802.11n
    Authentication         : WPA3-Personal
    Cipher                 : CCMP
    Connection mode        : Profile
    Receive rate (Mbps)    : 86.7
    Transmit rate (Mbps)   : 72.2
    Signal                 : 92%
    Rssi                   : -45
    Profile                : POCO X6 Neo 5G 2
    QoS MSCS Configured          : 0
    QoS Map Configured           : 0
    QoS Map Allowed by Policy    : 0
```